SIGMACAMP · ELECTROCHEMISTRY LAB · SESSION 4

Water Electrolysis

Experiment Packet

Audience: instructors and teaching assistants

Purpose: Bench instructions for instructors and TAs

Session hook: Predict a gas volume, then go and measure it

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

SESSION OVERVIEW & PLAN

Session 4 — Water Electrolysis and Gas Collection

Hook: Pure water barely conducts. Add one spoonful of salt-that-isn't-table-salt and the same water tears itself into hydrogen and oxygen — in a ratio you can measure with a ruler.

Learning outcomes

- Explain why pure and tap water electrolyze poorly (**too few mobile ions**)
- Show that **Na_2SO_4** raises current and therefore reaction rate, without being consumed
- Measure **current**, and use Faraday's law to **predict a gas volume before collecting it**
- Relate applied **voltage** to the ~1.23 V thermodynamic floor and to overpotential
- Collect H_2 and O_2 separately in the **U-tube** and confirm the **2:1** ratio
- Identify both gases by their **splint tests**, and hydrogen again by a balloon pop
- **Start** the electrolytic rust-removal cells that Session 5 opens with

Session plan (90 min)

Time	Activity	Reference
−5 min	**TA: switch on the balloon H_2 generator.** It must run the whole session — see the timing table in experiment.md Part D	preparation.md
0–8 min	Review Days 1–3; hook: two rods in pure water, nothing happens	lecture.md §1–3
8–18 min	Concepts: ions carry current, supporting electrolyte, 1.23 V, $\dot{n} = I/(nF)$, 2:1	lecture.md §3
18–26 min	**Quiz check + pre-lab prediction** (required before Part C)	lecture.md §4–5
26–36 min	**Part A** — Conductivity ladder: distilled → tap → electrolyte	experiment.md — Part A
36–50 min	**Part B** — Measure I; calculate $\dot{n}$; **write a volume prediction**	experiment.md — Part B
50–72 min	**Part C** — U-tube collection; ratio **and** Faraday check	experiment.md — Part C
72–82 min	**Part D** — Splint tests, then the balloon pop	experiment.md — Part D
82–90 min	**Part E** — Start derusting cells for Session 5	experiment.md — Part E

The one thing that ruins this session: starting the balloon generator when you reach Part D. At a realistic classroom current it needs **30–60 minutes** to fill. Start it before the students walk in.

Key reactions

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Overall:     2H₂O(l) → 2H₂(g) + O₂(g)

Cathode (−): 2H₂O + 2e⁻ → H₂ + 2OH⁻
Anode (+):   2H₂O → O₂ + 4H⁺ + 4e⁻     (acidic form)
          or 4OH⁻ → O₂ + 2H₂O + 4e⁻    (basic form)

Rate:        ṅ(H₂) = I/(2F)      ṅ(O₂) = I/(4F)
Rule of thumb: 1 ampere makes ≈ 7.5 mL of H₂ per minute at room conditions.
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Status

- [ ] Balloon H_2 generator tested and its fill time measured
- [ ] ~0.1 M Na_2SO_4 mixed and labeled **"NO SALT"**
- [ ] U-tube pilot-tested: fill level correct, ratio visible inside 20 min
- [ ] U-tube internal diameter measured and written on the board
- [ ] Splints and a lighter staged; balloon area cleared
- [ ] **No NaCl** anywhere in this room today except the sealed Session 5 corrosion jars
- [ ] Rusty nails, washing soda, and a **mains-powered supply** ready for Part E

PREPARATION

Session 4 — Preparation

The three things that decide whether this session works

1. **The balloon generator starts before the students arrive.** At a realistic classroom current a party balloon takes 30–60 minutes to fill. There is no way to hurry it.
2. **The U-tube is filled until liquid stands part-way up both side arms.** Filled any lower, the gas escapes out of the side arm and Part C silently collects nothing.
3. **The overnight derusting cells run on mains power, not a 9 V battery.** A 9 V alkaline is flat in about two hours, and Session 5 then opens on a nail that looks exactly like the control.

Everything else on this page is ordinary preparation.

Before class (instructor)

Pilot the measurements

- [ ] Run the **conductivity ladder** at a fixed voltage and record all three currents
- [ ] Work the **Faraday prediction** with your own pilot current — students will copy your method line for line
- [ ] Pilot the **U-tube**: confirm the fill level, and that a clear ratio appears inside 20 minutes
- [ ] **Measure the U-tube's internal diameter** and write it somewhere permanent
- [ ] Time the **balloon generator** at the current you intend to use, and record the minutes-to-usable in materials.md
- [ ] Rehearse both **splint tests** — the glowing-splint relight takes a little practice to get a splint to the right ember state

Solutions and hardware

- [ ] Mix **~0.1 M Na_2SO_4** stock; check whether your salt is anhydrous (3.6 g/250 mL) or **decahydrate** (8.1 g/250 mL)
- [ ] Label every container **"0.1 M Na_2SO_4 — NO SALT"**
- [ ] Pre-weigh Na_2SO_4 portions for the Part A rung-3 addition
- [ ] One jug of **distilled water** for the ladder's first rung
- [ ] Check graphite rods, stoppers (**and spares**), spade connectors, multimeters, fresh meter fuses
- [ ] Confirm every multimeter's mA range actually works — a blown fuse reads zero and looks exactly like "no reaction"

Part E staging

- [ ] Rusty nails — **two per group**, matched in rustiness, so the control is a fair comparison
- [ ] Washing soda, sacrificial anodes, open jars
- [ ] **Mains supply or USB bricks** confirmed and counted
- [ ] Overnight power policy confirmed **with the venue**, in writing if possible
- [ ] Ventilation confirmed for the room the cells will sit in overnight

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Day-of setup

- [ ] **T minus 5 minutes: switch on the balloon generator**
- [ ] Ladder stations: three labeled beakers, graphite, leads, series ammeter
- [ ] U-tubes filled to the correct level, side arms open, levels equal on both sides
- [ ] U-tube internal diameter written on the board
- [ ] Large sign: **"NO SALT IN ELECTROLYSIS — SODIUM SULFATE ONLY"**
- [ ] Balloon area cleared and marked; extinguisher present; long lighter staged
- [ ] Splints and collection tubes at each bench
- [ ] Part E station ready to go at the 82-minute mark, not assembled from scratch then
- [ ] Prediction sheets printed

Common failure modes

Issue	Prevention
"Part A is broken, nothing is happening"	Brief them first: a tiny current **is** the result
Meter reads zero after adding Na_2SO_4	Wrong range, not in series, blown fuse, or bad clip — check the fuse first
Ladder currents not comparable	Electrode spacing or immersion depth changed between rungs. Fix the geometry, fix the voltage
U-tube collects nothing	Filled below the side arms. The most common single failure in this session
Ratio slides toward 1 late in the run	H_2 level reached the side arm and is venting. Stop earlier
Poor ratios across the whole class	Nobody pre-saturated. Enforce the 2–3 min run before t = 0
Efficiency above 100%	Air trapped at t = 0, or a mis-measured diameter (it enters squared)
Balloon never fills	Started too late. Fall back to the splint tests and log the real fill time for next year
Students reach for the salt	Pre-brief; remove NaCl from the room entirely; only Na_2SO_4 is on the benches
Derusting nails look untouched next morning	9 V battery died. Use mains power

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Open questions / notes for next year

- Fixed voltage used for the ladder:
- Currents measured — distilled / tap / Na_2SO_4:
- Onset voltage for visible bubbling:
- U-tube internal diameter:
- Typical ratio and typical efficiency achieved by groups:
- Balloon generator current and actual fill time:
- Balloon demo permitted at this venue?
- Overnight power arrangement that was actually used:

MATERIALS CHECKLIST

Session 4 — Materials

Per group — Part A (conductivity ladder)

- [ ] Three beakers or clear cups (~200–400 mL), labeled 1 / 2 / 3
- [ ] Two **graphite** rods
- [ ] DC power supply able to hold a **fixed** voltage (or a battery pack)
- [ ] Multimeter for **current in series** (mA range)
- [ ] Way to read voltage (second meter, or the supply display)
- [ ] **Distilled water** — about 250 mL per group (a jug covers the class)
- [ ] Tap water
- [ ] Goggles

Per group — Part B (measure and predict)

- [ ] **Na_2SO_4** — 3.6 g anhydrous (or 8.1 g decahydrate) per 250 mL
- [ ] Calculator
- [ ] Printed prediction sheet — lecture.md §5 Part 3
- [ ] Stopwatch or phone

Per group / station — Part C (U-tube)

- [ ] Hoffman-style glass **U-tube** with side arms + acrylic stand
- [ ] Graphite electrodes with well-fitting rubber stoppers (**spares** — a split stopper ends the run)
- [ ] Red (+) / black (−) leads with spade connectors
- [ ] **~0.1 M Na_2SO_4**, enough to fill each U-tube with liquid standing in the side arms
- [ ] Ruler for column heights
- [ ] **Caliper or ruler to measure the tube's internal diameter** — needed once, for the whole class
- [ ] Marker or tape to mark the t = 0 levels

Part D — Gas identification

Splint tests (per group or per bench)

- [ ] **Wooden splints**
- [ ] Small test tubes (~10 mL) for collecting gas over solution
- [ ] Lighter or matches — TA-supervised

Balloon demo (instructor)

- [ ] Dedicated H_2 generator: beaker or bottle + two graphite electrodes + Na_2SO_4
- [ ] Stopper and tubing so **only cathode gas** reaches the balloon; anode vents to the room
- [ ] Power supply capable of the highest current you are willing to run

- [ ] **Small balloons**, pre-stretched by hand before use
- [ ] Long lighter or a candle on a stick
- [ ] Fire extinguisher accessible
- [ ] Cleared demo area, well away from every gas-generating cell

Part E — Start rust removal (end of class)

- [ ] Rusty iron nails or hardware, **2+ per group** (one treated, one control)
- [ ] Washing soda: **1 tbsp per 250 mL** warm water (preferred), or baking soda at the same rate — **never NaCl**
- [ ] Sacrificial steel strips or graphite rods
- [ ] Small jars or cups — **open, never sealed**
- [ ] **Mains DC supply or 5 V USB power brick** — see the note below
- [ ] Labels and a camera for "before" photos

Power source for the overnight cells

Option	Verdict
Mains DC supply, 5–12 V, current-limited to ~0.5 A/cell	**Best**
5 V USB power brick	Good, cheap, self-limiting
9 V alkaline, 60–90 min during class	Acceptable fallback
9 V alkaline overnight	**Will not work** — flat in about two hours

Explicitly excluded

- [ ] ~~Table salt (NaCl)~~ — **not in any bath in this room today**

Optional / next year

- [ ] Conductivity meter — turns the Part A ladder quantitative (µS/cm)
- [ ] Universal indicator — makes OH^- and H^+ production visible at the electrodes; the best cheap upgrade to this session
- [ ] Thermometer, for the vapour-pressure correction with strong students

Pilot notes — fill these in, they are next year's most valuable page

Parameter	Value from pilot
Fixed voltage used for the ladder	_______ V
I, distilled water	_______ mA
I, tap water	_______ mA
I, 0.1 M Na_2SO_4	_______ mA
Onset voltage for visible bubbling	_______ V

Parameter	Value from pilot
U-tube internal diameter	_______ cm
Typical $V(H_2)/V(O_2)$	_______
Typical current efficiency (measured ÷ predicted)	_______ %
Balloon generator current	_______ A
Minutes to a usable balloon	_______ min
Balloon demo permitted at this venue?	

EXPERIMENT PROCEDURE

Session 4 — Experiment: Water Electrolysis

> **Instructors:** conductivity ladder, U-tube fill level, why the ratio runs high, balloon timing — instructor.md.

Session arc

1. **Part A** — Conductivity ladder: distilled → tap → electrolyte (why nothing happens without ions)
2. **Part B** — Measure **current**; use Faraday's law to **predict** a gas volume
3. **Part C** — Collect the gases in the **U-tube**; check the **2:1 ratio** *and* your prediction
4. **Part D** — Identify both gases: splint tests, then the **H_2 balloon pop**
5. **Part E** — Start the overnight **rust-removal** cells for Session 5

Before anything else: start the balloon generator

The balloon demo in Part D is the only part of this session that cannot be rushed, because the gas has to be *made*, one electron at a time. Faraday sets the pace and there is no way around it:

Current	H_2 produced	Time to fill 100 mL	250 mL	500 mL
0.5 A	3.8 mL/min	26 min	66 min	132 min
1.0 A	7.6 mL/min	13 min	33 min	66 min
2.0 A	15.2 mL/min	7 min	16 min	33 min
3.0 A	22.8 mL/min	4 min	11 min	22 min

A party balloon inflated to a **modest 10 cm across holds about 500 mL**. Read across that row and it is obvious: **switch the generator on before the students arrive** and let it run through Parts A–C. See Part D for the build.

Electrolyte — sodium sulfate (Na_2SO_4)

Never use table salt (NaCl). Chloride is oxidized at the anode in preference to water and produces **chlorine gas**.

Recipe — ~0.1 M Na_2SO_4

Batch	Anhydrous Na_2SO_4	Decahydrate (Glauber's salt)	Water
250 mL	**3.6 g**	**8.1 g**	Fill to 250 mL
500 mL	**7.1 g**	**16.1 g**	Fill to 500 mL
1.0 L	**14.2 g**	**32.2 g**	Fill to 1.0 L

Check which one you bought — the decahydrate is more than half water by mass, and weighing it as if it were anhydrous gives you a bath at less than half the intended strength.

Stir until fully dissolved. Label **"0.1 M Na_2SO_4 electrolyte — NO SALT"**.

Water: tap is fine. Distilled is needed only for the first rung of the Part A ladder.

Backup if Na_2SO_4 is unavailable: baking soda, 1–2 tsp per 500 mL. It works, but it fizzes CO_2 at the anode and muddies the Part C ratio, so it is a fallback and not an equal substitute.

Part A — The conductivity ladder (26–36 min)

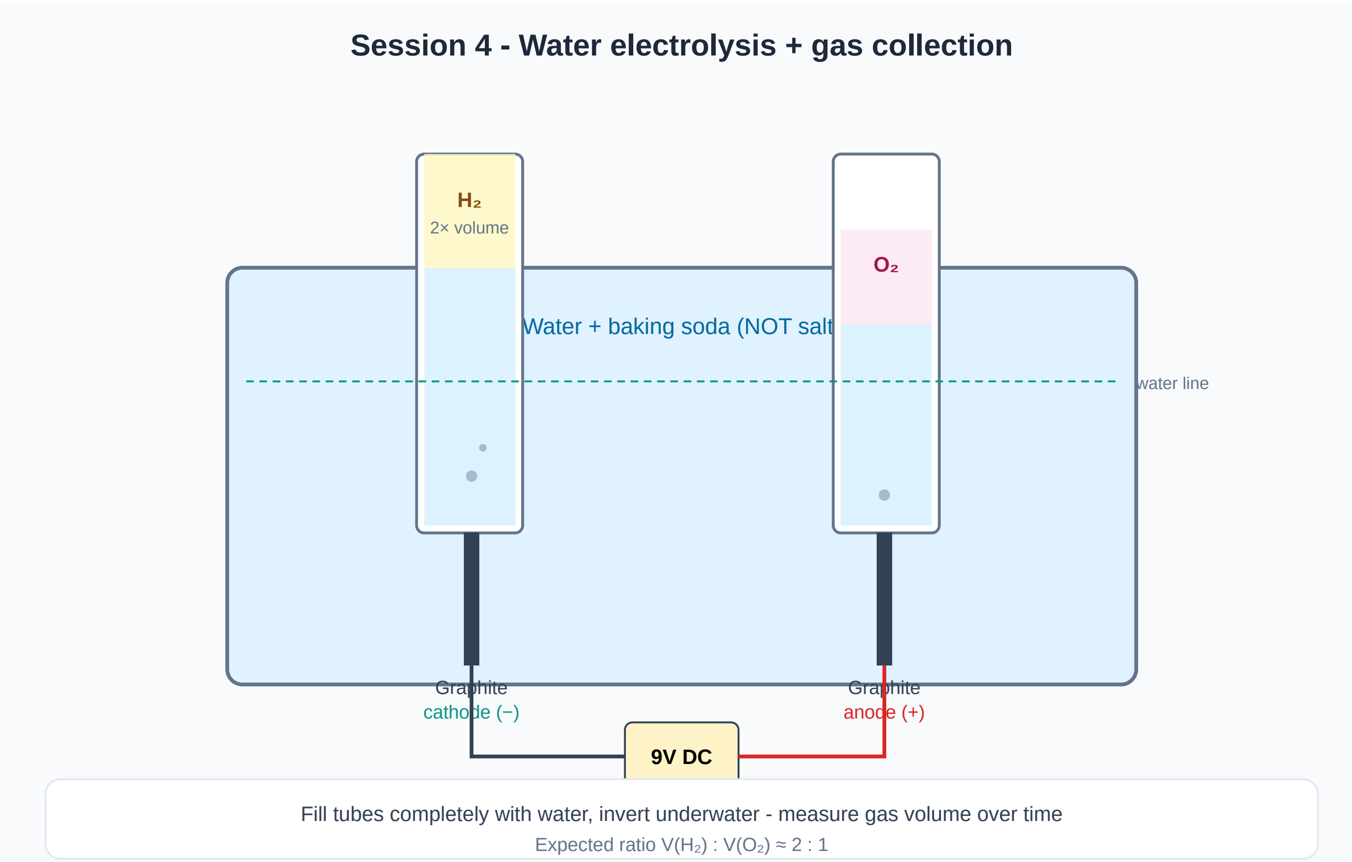

Figure 1 — H_2 at the cathode (−), O_2 at the anode (+). With no ions in solution, almost no current flows and almost nothing happens.

The point of this part

Last session students used electricity as a **measuring stick**. Today they first have to earn the current. Three liquids, **one fixed voltage**, one meter — the only thing that changes is what is dissolved in the water.

Materials (beaker cell)

- Beaker or clear cup (~200–400 mL) — ideally three, one per rung
- Two **graphite** rods
- DC supply set to a **fixed voltage** (5 V is a good choice) — do not change it between rungs
- Multimeter on DC **mA**, wired **in series**
- Distilled water, tap water, and Na_2SO_4
- Goggles

Procedure

1. Set the supply to your chosen voltage and **write it on the board**. It does not change for the rest of Part A.
2. Clamp the two graphite rods **2–3 cm apart**, immersed to the same depth every time. Spacing and depth are part of the experiment — changing them invalidates the comparison.
3. **Rung 1 — distilled water.** Power on for 2 minutes. Record current. Look hard for bubbles.
4. **Rung 2 — tap water.** Same electrodes, same spacing, same voltage. Record current.
5. **Rung 3 — add Na_2SO_4** to the tap water to make roughly 0.1 M (3.6 g per 250 mL), stir until dissolved, and record current again.

Data — the ladder

Rung	Liquid	Applied V	Current I (mA)	Bubbles?
1	Distilled water			
2	Tap water			
3	Tap water + 0.1 M Na_2SO_4			

Current in rung 3 ÷ current in rung 1 = _______

What to expect

Distilled water gives a current so small that many meters read zero — pure water contains only about **one ion pair in every 550 million molecules**. Tap water usually gives something small but *real*, a few mA, because it carries dissolved minerals; this rung is worth doing precisely because it is not zero. The electrolyte rung is typically **hundreds of times larger** than rung 1.

Say out loud: *the water did not change between rung 2 and rung 3.* The same H_2O molecules are being split. All that changed is that charge now has a way to cross the gap.

Talking point: *Water is the reactant. The electrolyte is the road. Without a road, the reactant never gets to the electrode.*

Part B — Measure the current, predict the gas (36–50 min)

This part produces a **number that Part C will test**. Do not skip ahead to the U-tube.

Procedure

1. Keep the rung-3 beaker running at your fixed voltage with the ammeter in series.
2. Let the current settle (10–30 s) and record it. If it drifts, record a high and a low and use the average.
3. Watch both electrodes. **One side bubbles visibly faster.** Record which — this is a prediction you will check in Part C.
4. Optional, if the supply is adjustable: wind the voltage down until bubbling just stops, and record that threshold.

Data — with Na_2SO_4

Quantity	Value
Applied voltage V (V)	

Quantity	Value
Current I (mA) → in amperes, I = _______ A	
Which electrode bubbles faster, (+) or (−)?	
Voltage at which bubbling just disappears (optional)	

The two voltages worth naming

- **1.23 V** is the thermodynamic floor: below it, splitting water is uphill no matter how patient you are.
- Real cells need **more** — typically **1.6–2.0 V before anything is visible**, and several volts to run briskly. The excess pays for **overpotential** (the electrode surface is slow at making gas) and for the resistance of the solution.
- Students record the **applied V** they actually used, not the textbook number.

If you hunted for the bubbling threshold above, that measured onset is the 1.23 V floor *plus* overpotential — you have measured a real irreversibility, not an error.

Rate from current (Faraday, again)

Same law as Session 3, new products. Hydrogen needs **2 e^- per molecule**, oxygen needs **4**:

```
ṅ(H₂) = I / (2F)          ṅ(O₂) = I / (4F)          F = 96,485 C/mol e⁻
```

So for the same current, H_2 comes off at **exactly twice** the mole rate of O_2 — the 2:1 ratio is already there in the electron bookkeeping, before you collect a single bubble.

Converting to a volume at room conditions (25 °C, 1 atm), where 1 mol of any gas occupies **24.45 L**:

```
V̇(H₂) in mL/min = [ I / (2F) ] × 24,450 × 60 ≈ 7.6 × I      (I in amperes)
```

Rule of thumb to put on the board: one amp makes about 7.5 mL of hydrogen per minute.

Prediction sheet — fill in before you touch the U-tube

Step	Expression	Your value
Measured current I	_______ A	
$\dot{n}(H_2) = I/(2F)$	_______ mol/s	
$\dot{n}(O_2) = I/(4F)$	_______ mol/s	
H_2 volume rate ≈ 7.6 × I	_______ mL/min	
Predicted H_2 after 10 min	_______ mL	
Predicted O_2 after 10 min	_______ mL	

Check before moving on: your two mole rates must differ by exactly a factor of 2. If they do not, you divided by the wrong number of electrons.

Unit trap: the meter says mA. 300 mA is **0.300 A**. Getting this wrong makes you wrong by a factor of a thousand, and a thousandfold error looks perfectly reasonable on a calculator screen.

Part C — Collect the gases and test the prediction (50–72 min)

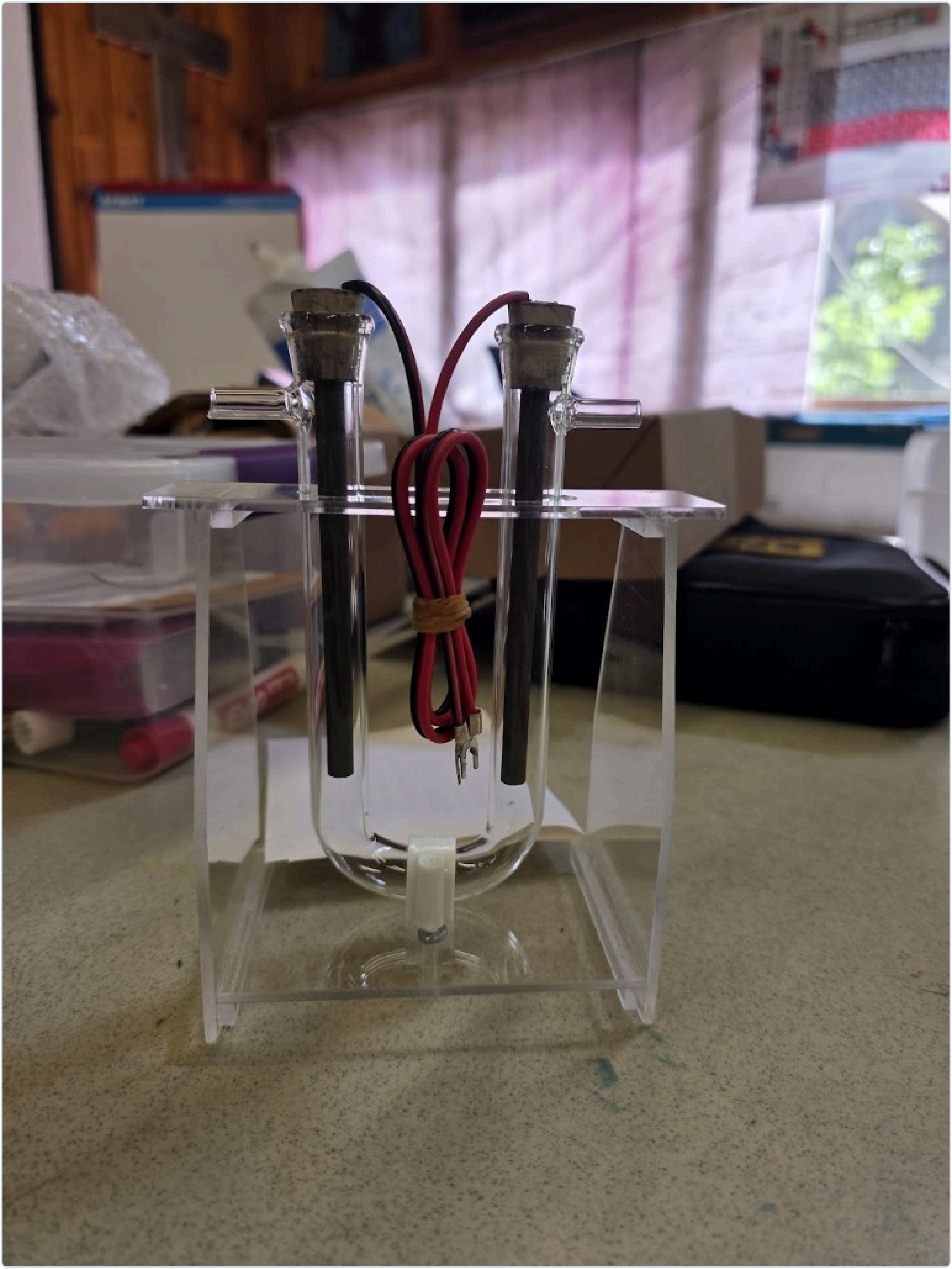

Figure 2 — The U-tube. Each limb is closed at the top by a stopper carrying a graphite rod. Each limb has an open side arm near the top. Gas gathers in the sealed dome under each stopper and pushes the liquid level down.

How this apparatus actually works — read before filling

This is the step that most often fails, and it fails for one reason: **the fill level**.

Gas is trapped in the closed space between the stopper and the liquid surface. As gas accumulates, the liquid in that limb is pushed **down**, and the displaced liquid rises up the open side arm. That side arm is the vent — it is what lets the liquid move. It must stay open.

The consequence is the rule that decides whether Part C works:

> **Fill until the liquid stands part-way up both side arms.** If you fill only to *below* the side-arm junction, the gas escapes straight out of the side arm and you will collect nothing while apparently doing everything right.

And the matching stop condition:

> **Stop the run before the falling liquid level in the H_2 limb reaches the side-arm junction.** Past that point, hydrogen vents and your ratio silently collapses toward 1.

The H_2 limb fills twice as fast, so it always hits the junction first. It sets the length of your run.

Materials

- Hoffman-style glass **U-tube** with side arms + acrylic stand
- Graphite electrodes + rubber stoppers (a snug seal is essential — a leaking stopper is indistinguishable from bad chemistry)
- Red (+) / black (−) leads with spade connectors
- DC supply
- **0.1 M Na_2SO_4**, enough to fill the U-tube
- Ruler, and a **caliper or ruler to measure the tube's internal diameter**
- Goggles

Build and run

1. Fill the U-tube with Na_2SO_4 until the liquid stands **part-way up both side arms**, at the **same height on both sides**.
2. Insert the stoppered graphite electrodes and press them home. Leave both side arms **open**.
3. Connect **black (−) to one limb, red (+) to the other**. Note on your sheet which limb is which.
4. **Power on and run for 2–3 minutes before you start timing.** This is not wasted time — it saturates the solution with dissolved gas so that the bubbles you count from now on actually stay in the tube. Skipping it is the single biggest source of a bad ratio.
5. Now mark both liquid levels. This is **t = 0**.
6. Every 3 minutes, record how far the level has **fallen** in each limb.

Gas log

Measure the **drop in liquid level** in each limb, in cm.

Time (min)	Drop in (−) limb, cm	Drop in (+) limb, cm	Ratio
0	0	0	—
3			
6			
9			
12			
15			

Both limbs have the same bore, so for the *ratio* you never need the diameter — centimetres of column are as good as millilitres. Convert to volume only for the Faraday check below.

Result 1 — the stoichiometric ratio

$$2H_2O(l) \rightarrow 2H_2(g) + O_2(g) \quad \Rightarrow \quad V(H_2) : V(O_2) = 2 : 1$$

The faster limb must be hydrogen. **Check that it is the one wired to (−)**; if it is not, your leads are swapped.

Honest expectation: 2.0 to 2.6, with oxygen running short. A ratio slightly above 2 is the normal, correct result, for three reasons worth telling students:

Cause	Effect on the ratio
O_2 is about twice as soluble in water as H_2	Some oxygen dissolves instead of collecting → ratio high
The graphite anode is slowly attacked, and some oxygen ends up as CO_2, which is very soluble	Ratio high
Oxygen evolution has the larger overpotential and is slower to get started	Ratio high early in the run

A ratio **below 2** is the one that signals a fault, not a subtlety: suspect a leaking stopper on the H_2 side, or that the level in the H_2 limb reached the side arm and gas vented.

Result 2 — the Faraday check (this is the real payoff)

Convert one of your column drops into a volume and compare it with the prediction you wrote in Part B.

$$\text{Volume (mL)} = \pi \times (d/2)^2 \times h \qquad d = \text{tube internal diameter in cm, } h = \text{drop in cm}$$

Tube internal diameter	1 cm of column equals
1.5 cm	1.77 mL
2.0 cm	3.14 mL
2.5 cm	4.91 mL

Quantity	Value
Predicted H_2 volume from Part B	_______ mL
Measured H_2 volume	_______ mL
Measured ÷ predicted (× 100%)	_______ %

This percentage is the **current efficiency** — the same idea students met in Session 3, but this time they can actually check it, because gas is far easier to measure than a sub-micrometre film of silver.

Expect **70–95%**. Losses go to dissolved gas, leaks, and side reactions on the graphite. Getting *more* than 100% means a measurement problem: an air bubble trapped at t = 0, or a mis-measured tube diameter.

Pre-run checklist

- [ ] Electrolyte is **Na_2SO_4** — not salt
- [ ] Liquid stands **part-way up both side arms**, level on both sides
- [ ] Side arms **open**; stoppers seated firmly
- [ ] Ran 2–3 min *before* marking t = 0
- [ ] Polarity recorded; tube diameter measured
- [ ] Goggles on

Part D — Identify the gases (72–82 min)

Two tests. The splint tests are quick, reliable, and done first. The balloon is the finale.

D1 — Splint tests (2 minutes, works every time)

The chemistry here is a genuine identification, not a party trick: **hydrogen burns, oxygen makes other things burn.**

Gas	Test	Result
H_2 (from the cathode, −)	Hold a **lit** splint at the mouth of the tube	Sharp **squeaky pop** as it burns in air
O_2 (from the anode, +)	Blow out a splint so it is **glowing**, then insert it	The glowing ember **relights**

Collecting the gas: fill a small test tube with electrolyte, invert it over one electrode in a beaker, and let bubbles displace the liquid. A few millilitres is plenty. At a modest 0.3 A, hydrogen fills a 10 mL tube in about **4 minutes** — set these up at the start of Part C and they will be ready now.

Keep the tube **mouth-down** until the moment of the test. Hydrogen is far lighter than air and leaves an upright tube immediately.

D2 — The hydrogen balloon (instructor only)

This is a demo. Students watch. Only the instructor handles flame.

Setup — running since before class started:

- A dedicated cell: beaker or bottle of Na_2SO_4 with two graphite electrodes
- A stopper and tube arranged so that **only cathode gas** reaches the balloon; the anode vents freely to the room

- The highest current you can run comfortably — see the timing table at the top of this file
- **Pre-stretch the balloon** by hand before attaching it. A fresh balloon is stiff and will resist the very low pressure this cell can generate.

Procedure:

1. When the balloon is modestly inflated, pinch it off, remove it, and **tie it**.
2. Carry it **well away** from the electrolysis cell — that cell is still making hydrogen.
3. Clear the area. Warn the room that the bang is loud and sudden.
4. Ignite with a **long** lighter or a candle on a stick, at arm's length.
5. One sharp bark. That is 500 mL of hydrogen meeting the oxygen in the room.

Safety — non-negotiable:

- [ ] Instructor is the only person near the flame
- [ ] Balloon contains **cathode gas only** — never a deliberate H_2/O_2 mixture, which detonates rather than burns
- [ ] Never in a rigid or sealed container. A balloon deforms; glass shatters
- [ ] Fire extinguisher present; nothing flammable overhead
- [ ] Everyone in goggles; warn anyone sensitive to noise
- [ ] Skip entirely if the venue forbids open flame — D1 already proved the chemistry

Where the energy came from

Ask immediately after the pop, while the room is still loud: *that bang was the energy we spent all session pushing into the water, coming back out in a quarter of a second.* Hydrogen did not create energy. It **stored** the energy the power supply put in — that is what "energy carrier" means, and it is why hydrogen is discussed as a way to move renewable electricity around rather than as a fuel we dig up.

Part E — Start electrolytic rust removal (82–90 min)

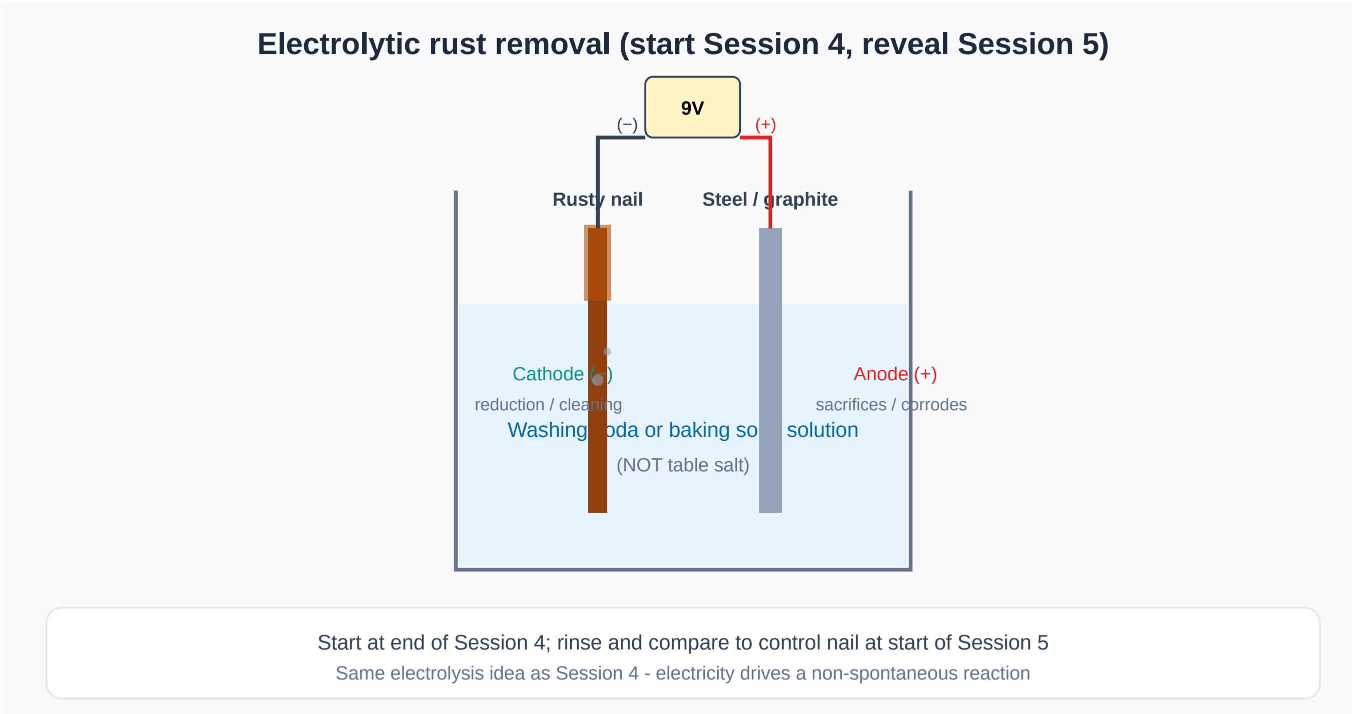

Figure 3 — Start these cells now; they are the first thing students see in Session 5.

Power supply — read this before wiring

Do not run these overnight on a 9 V alkaline battery. A derusting cell draws a few hundred milliamps, and a 9 V alkaline holds roughly 500 mAh. It will be flat within about two hours, warm, and possibly leaking — and Session 5 opens on a nail that looks exactly like the control.

Option	Verdict
Mains DC supply, 5–12 V, current-limited	**Best.** Set the limit to ~0.5 A per cell
USB power brick at 5 V	Good, cheap, and self-limiting. Slower but perfectly adequate
9 V alkaline for a 60–90 min in-class run	Acceptable if no mains option exists — but then run it during class, not overnight
9 V alkaline left overnight	Will not work

Derusting electrolyte

Ingredient	Amount
Washing soda (Na_2CO_3) — preferred	**1 tbsp (~15 g) per 250 mL** warm water
or baking soda ($NaHCO_3$)	**1 tbsp per 250 mL** warm water

Never NaCl. Chloride at the anode makes chlorine. Keep this bath physically separate from the Na_2SO_4 electrolysis benches.

Setup steps

1. Dissolve the washing soda in warm water; pour into a jar.
2. **Rusty iron → (−) cathode.** Say it aloud as a class. The polarity is the whole experiment, and reversing it dissolves the nail instead of cleaning it.
3. **Sacrificial steel or graphite → (+) anode.** Position it facing the rusty surface, without touching.
4. Confirm gentle bubbling at both electrodes within a few seconds.
5. Label each jar. **Photograph the "before".** Without that photo, tomorrow's comparison is an argument.
6. Set up a **control**: an equally rusty nail in the same solution with **no power**. This is what makes the reveal a result rather than an anecdote.

Overnight safety

- [ ] Cells are in an **open** jar in a **ventilated** room — this cell makes hydrogen all night, and it must never be sealed
- [ ] No ignition sources nearby
- [ ] Wiring secured so nothing can short or tip
- [ ] Venue's overnight-power policy confirmed **in advance**

End-of-class checklist

- [ ] Every cell bubbling, polarity double-checked by a TA
- [ ] Control jar labeled and unpowered
- [ ] Before photos taken
- [ ] Room ventilated, jars open, power secured

Troubleshooting

Problem	Cause	Fix
Rung 1 reads exactly zero	Correct, and it is the lesson	Check the meter works by touching the leads together
Tap water gives more current than expected	Hard water; normal	Keep it — it makes the ladder more interesting, not less
No bubbles after adding Na_2SO_4	Bad clip, wrong meter range, undissolved solid	Check contacts; stir; confirm the meter is in series
Chlorine smell	**Salt was used**	Stop, ventilate, dump the bath, remake with Na_2SO_4
U-tube collects no gas at all	**Filled below the side arms** — gas is venting	Refill until liquid stands up in both side arms
Ratio drifting toward 1 late in the run	H_2 level reached the side arm and is venting	Stop the run earlier; use the last good reading
Ratio well below 2	Leaking stopper on the H_2 limb	Reseat stoppers; restart
Ratio 2.2–2.5	Not a fault — this is the expected result	Discuss solubility and anode side reactions

Problem	Cause	Fix
Measured volume far below prediction	Late t = 0, leak, or a mis-measured diameter	Recheck diameter; 70–95% is the normal window
Balloon barely inflating	Started too late, current too low, or balloon too stiff	Nothing to be done today — run D1 instead and note the fill time for next year
Graphite crumbling, bath going grey	High current over a long run	Expected on the balloon cell; lower the current, replace rods

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Safety checklist

- [ ] **No NaCl** in any bath in this room today
- [ ] Ammeter in **series** — never clipped across the supply
- [ ] Goggles for all wet work
- [ ] U-tube side arms **open** — never run this apparatus sealed
- [ ] Flame handled by the instructor only, away from every gas-generating cell
- [ ] Overnight derusting jars open and ventilated

Experiment status

- [ ] Conductivity ladder piloted; all three currents recorded at a fixed V
- [ ] Faraday prediction arithmetic worked through with the pilot current
- [ ] U-tube fill level verified — liquid standing in the side arms
- [ ] Tube internal diameter measured and recorded
- [ ] Splint tests rehearsed (both gases)
- [ ] Balloon generator timed — minutes to a usable balloon: _______
- [ ] Derusting cells staged with a **mains** power source

SHARED SAFETY NOTES

Syllabus Safety Notes

Apply these across all sessions. Session-specific hazards are listed in each session folder.

The three rules that matter most

1. **Salt belongs in exactly one place this week.** NaCl is used only for the Session 5 corrosion jars, where nothing is driven hard enough to make chlorine. It is **forbidden** in every electrolysis bath and in the derusting cells, where chloride is oxidized at the anode to **chlorine gas**. This is a competing half-reaction, not an arbitrary rule.
2. **Any cell left running unattended must be open and ventilated.** Electrolysis and derusting cells evolve hydrogen continuously. An overnight derusting cell can generate a couple of litres of it. Open jars, ventilated room, no ignition sources, nothing sealed. Ever.
3. **Disconnect power before touching anything wet.** Every time, as a ritual, especially at the Session 5 reveal.

General rules

1. **No eating experimental materials.** Fruit, potatoes and liquids used in experiments are no longer food, and this includes the ones that started out edible.
2. **Ammeters go in series.** A meter on a current range clipped across a supply is a short circuit and will blow its fuse at best.
3. **Avoid short circuits generally.** Batteries heat up fast when shorted.
4. **Collect metal-salt waste.** Copper and silver solutions are not poured down the drain.
5. **Water choice:** tap water is fine for most baths. **Distilled is required for the Session 3 silver bath**, so chloride does not precipitate silver, and is worth having for the first rung of the Session 4 conductivity ladder.
6. **Goggles for all wet work.** Gloves additionally for silver nitrate.
7. **Rusty hardware is sharp** and often outdoors-dirty. Forceps or gloves; no bare-handed scrubbing.

Session-specific hazards

Session	Key hazard	Mitigation
1	Sharp nails and wires	Supervise handling; wash hands after handling metals
2	Copper sulfate solution	Goggles; no ingestion; collect waste
3	**Silver nitrate**	Gloves **and** goggles; stains skin and clothing near-permanently; small volumes (30–40 mL); collect all waste; **no ammonia, ever**
4	Gas evolution; hydrogen ignition	No NaCl; ammeter in series; U-tube side arms open; balloon and flame **instructor only**
5	Corrosion demos; overnight derusting	Washing soda only, never NaCl; disconnect before handling; jars open and ventilated overnight

Silver nitrate and ammonia — Session 3

Never add ammonia, or any ammonia-containing cleaner, to a silver nitrate solution. Ammoniacal silver solutions (Tollens' reagent) deposit **silver nitride and related compounds** on standing, which are friction- and shock-sensitive explosives. The hazard is not the fresh solution — it is any residue left in a bottle, on a stirrer, or around a cap.

This matters because the "silver mirror" recipes students will find online use exactly this chemistry. Say so out loud during the Session 3 safety brief, before someone tries it at home.

It also does not even work for our purpose: ammonia complexes copper about as strongly as it complexes silver, so it does nothing to suppress the displacement reaction we are trying to control.

Hydrogen — Session 4, instructor only

The balloon demo is optional. The chemistry is fully demonstrated by the splint tests without any of this risk, so skip the balloon entirely if the venue forbids open flame or if you are at all unsure.

If you do run it:

- Feed **cathode (H_2) gas only** into the balloon. The anode vents separately to the room.
- **Never a deliberate H_2/O_2 mixture.** Pure hydrogen in air *deflagrates* — it burns at the surface and gives a bark. A stoichiometric mixture **detonates**, which is a different event entirely and not a classroom demonstration.
- **Never a rigid or sealed container.** A balloon tears; glass becomes shrapnel.
- Modest inflation only.
- Tie off, then carry it **well away** from the electrolysis cell, which is still making hydrogen.
- Long lighter at arm's length; area cleared; extinguisher present and its location known.
- Warn the room. The bang is loud, sudden, and some students find it distressing.

Splint tests — the low-risk alternative

Gas	Test	Result
H_2	Lit splint at the mouth of the tube	Squeaky pop
O_2	Glowing (not flaming) splint inserted	Ember relights

A few millilitres of gas in a small test tube is plenty. Keep hydrogen tubes **mouth-down** until the moment of testing. TA-supervised; students may hold the tube, an adult holds the flame.

Waste

Waste	Disposal
Copper sulfate solution	Labeled waste container; not down the drain
Silver nitrate solution and rinses	Labeled waste container; not down the drain. Drop a coil of bare copper wire in — it strips the silver back out as metal for recovery
Sodium sulfate electrolyte	Drain is acceptable; dilute well

Waste	Disposal
Washing soda derusting sludge	Let solids settle, bag the sludge, dilute the liquid to drain
Saltwater corrosion jars	Drain; dispose of rusty metal in solid waste